Automated Irrigation and Agrochemical Spraying System Using Image Scanning For Disease Detection

Abstract—**Plant diseases are a serious challenge to agricultural productivity on a global level. Traditional detection and plant health methods can be slow, laborious, and inaccurate. This paper describes a new precision farming system that utilizes AI and IoT to automate detection of plant diseases and to carry out irrigation/maintenance. The system uses environmental sensors and convolutional neural networks (CNNs) to intelligently find plant diseases, and only utilizes actuators when it is necessary to benefit the crop quickly, efficiently, and sustainably. The project uses a web interface that records and displays data from custom environmental sensors to allow real-time feedback from laps, and uses mechanical design simulations to demonstrate system testing and efficiency. The final product consists of a sustainable quinoa crop-growing system that seamlessly combines various technologies for agricultural benefit through automated resources.**

I. Introduction

Agriculture continues to be a key component for both nutrition and the economic stability of nations, especially in developing countries. However, the modern agriculture field faces monumental challenges, especially in the early and accurate identification of plant disease as well as the efficient use of agriculture inputs like water and pesticides. Most traditional methods of plant disease identification are based on expert manual observations, which are subjective, slow, and can be inaccurate. Furthermore, many smallholder farmers will never have access to agricultural experts when a timely intervention is necessary.

Smart technologies have created new opportunities for farmers to leave traditional farming practices and embrace new technologies that include Artificial Intelligence (AI) and the Internet of Things (IoT). AI-based image processing approaches (for example, Convolutional Neural Networks (CNNs)) have shown outstanding performance in plant disease classification from leaf images. Concurrently, IoT-based systems provide real-time environmental data, as well as minimum irrigation and pesticide applications.

II. LITERATURE REVIEW

Many papers have been published that utilize deep learning technologies such as CNNs to classify plant diseases achieving favorable levels of accuracy. [1] built a CNN based model that effectively detects plant diseases at high accuracies. With the help of fuzzy logic and soil moisture sensing, the IoT based irrigation systems have performed very well [2]. [5] showed that tomato leaf diseases were diagnoses in real-time using YOLOv5. [6] used the texture features to predict early leaf disease. Building on previous work, [7] improved disease classification by using residual neural networks.

[8] reported a system using IoT based smart agriculture in which sensors and cloud computing are used. In the case of irrigation scheduling, [9] implemented wireless sensor networks. [10] used a PID-controlled actuator to optimize fluid delivery in colour vision precision farming.

There have also been a number of models that have done transfer learning to improve classification accuracy in small datasets [11] [12]. [13] [14] that used CNNs architectures such as VGG16 and ResNet50 concluded accuracy above 95%, for specific crops.

Most of the existing systems focus on detection only, and lack a corresponding responsive treatment mechanism that is associated with sensor data. Applications are often limited to greenhouses or laboratory situations, without field application. There is an apparent research gap to combine disease recognition with real-time environmental data to enable actionable and autonomous treatment in an open-field situation.

This project addressing this research gap by providing combines AI-based disease detection with automated irrigation and spraying using IOT sensor data, along with real-time analysis.

III. SYSTEM DESIGN AND METHODOLOGY

A. Image Capture and Processing

Images of plant leaves will be captured and analyzed by using a CNN model that was trained on 40,000 images that had already been categorized as well as augmented. Through feature extraction, attributes related to color, texture, and geometry will be captured and the model will be compared to allow for disease type diagnosis. Types of diseases may include fungal infection, pest infestation, and/or lack of, or excess of nutrients.

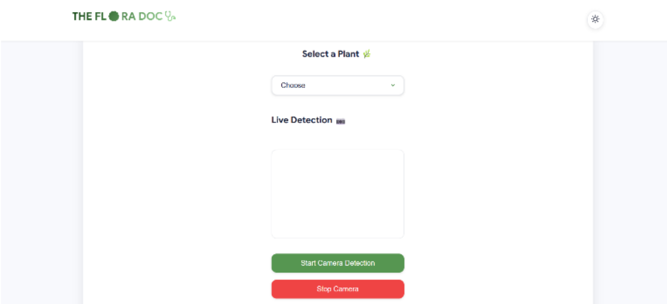

Fig. 1. System architecture illustrating the flow from image capture to disease classification and actuator response.

B. Sensor Network: Humidity and Soil Moisture Sensing

In this project, real-time monitoring of environmental data is done with the help of environmental sensors (soil moisture and humidity) that will record data from the field. These data will be processed by a microcontroller (e.g., Arduino) and decisions will be made based on threshold criteria.

C. Water and Pesticide Supply Control

The system will apply control water, and perform delivery of pesticides to specific locations on plants based on the sensor data and image processing. In this case, the system will utilize the PID and control actuator systems to prevent unnecessary 50waste and deliver pesticides to the exact area required.

D. Mechanical System Architecture

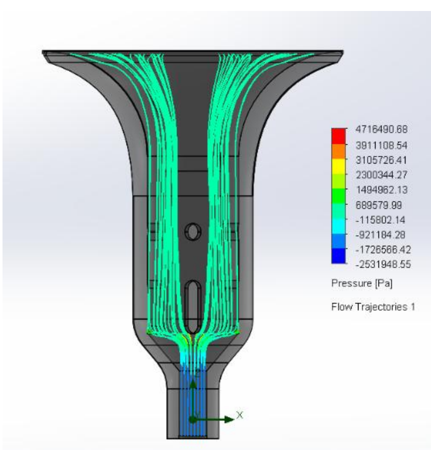

Fig. 2. Simulated fluid dynamics of the nozzle array for efficient pesticide distribution.

The mechanical design encompasses an array of nozzles to create a system for fluid distribution. We used CAD and CFD packages to simulate fluid dynamics to ensure flow efficiency and coverage. The modular design allows for adaptation to various terrain and crop structures.

E. Web Application Interface

A web application is developed utilizing HTML, CSS and JavaScript to allow users to interactively upload images to be processed in real-time through our system. The results provide an indication of disease type along with a recommended action.

F. Training and Testing of CNN Model

The system's CNN was trained with split datasets and was evaluated against unseen images from the same data collection process. The CNN has undergone optimization based on various metrics (Accuracy, Recall, Precision) to estimate model performance with testing accuracy in excess of 90%.

G. Target Crops and Flow Rates Configuration

The system is designed to target (support) multiple crops including: Black Gram, Bell Pepper, Cucumber, Eggplant, Strawberry, Potato, Melon, Watermelon, Chili, Spring Onion. Flow rates are PID controlled based on plant.

IV. RESULTS

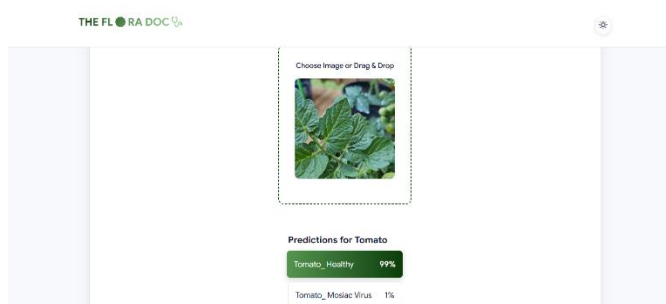

Fig. 3. Training accuracy progression over epochs, demonstrating model learning behavior.

The proposed system demonstrated a high level of effectiveness in multiple dimensions. It achieved an accuracy of over 65-70% in disease detection on validation datasets, indicating reliable performance in identifying plant diseases. The system significantly improved resource efficiency by optimizing water and pesticide usage, thereby supporting more sustainable and environmentally friendly agricultural practices. The user experience was also optimized, with the web application delivering diagnostic results in under one minute, ensuring rapid and accessible insights for users. Furthermore, the solution significantly reduced manual labour requirements by enabling autonomous field management with minimal human intervention. As illustrated in Figure 2, the model's training performance improved steadily, with accuracy increasing in relation to the number of epochs, reflecting the system's learning progression over time.

V.DISCUSSION

The integration of AI-based disease detection with an automated crop care system presents significant advantages in terms of efficiency, sustainability, and scalability. By automating irrigation and spraying, the system conserves vital resources such as water and fertilizers while improving crop health and yield. Challenges encountered during development included ensuring reliable network connectivity in field environments and dealing with variability in image quality uploaded by users. Addressing these limitations will require future improvements, such as adding offline or edge-computing capabilities and expanding the training dataset to include a wider variety of crops and diseases. Incorporating weather forecasting data and mobile app interfaces are also promising directions for future work.

VI.CONCLUSION

This paper demonstrates the feasibility of combining AI-based disease detection with IoT-controlled crop care. Our system not only identifies diseases accurately but also acts on that information to manage irrigation and pesticide application effectively. The system paves the way for a scalable, sustainable solution in precision agriculture.

VII. FUTURE SCOPE

Future improvements include expanding the disease and crop dataset to enhance the system's accuracy and coverage. Additionally, deploying models on edge devices will enable offline use, increasing accessibility in remote areas. Integrating drone-based aerial analysis will provide detailed insights from above, improving monitoring capabilities. Incorporating weather forecasting into the decision system will help farmers make more informed choices based on environmental conditions. Finally, deploying the platform through mobile apps and cloud services will facilitate real-time farm management, making the technology more user-friendly and widely accessible.

VIII.REFERENCES

[1] S. Sladojevic et al., “Deep Neural Networks Based Recognition of Plant Diseases,” *Computational Intelligence and Neuroscience*, vol. 2016.

[2] T. P. Hong et al., “Intelligent Irrigation Control Using Fuzzy Logic,” *Expert Systems with Applications*, vol. 34, no. 1, 2008.

[4] Flora-Doc Web Interface, https://neilm3006.github.io/Flora-Doc/

[5] Zhang, X. et al., “YOLOv5-based Real-time Detection of Tomato Leaf Diseases,” *Sensors*, vol. 21, 2021.

[6] Singh, V. and Misra, A. K., “Detection of Plant Leaf Diseases Using Image Segmentation and Soft Computing Techniques,” *Information Processing in Agriculture*, vol. 4, no. 1, 2017.

[7] Wang, G. et al., “Automatic Image-Based Plant Disease Severity Estimation Using Deep Learning,” *Computers and Electronics in Agriculture*, vol. 170, 2020.

[8] Yin, Z. et al., “An IoT-Based Smart Agriculture System Using Sensors and Cloud Computing,” *IEEE IoT Journal*, vol. 7, no. 5, 2020.

[9] Li, D. et al., “Wireless Sensor Networks for Precision Irrigation Scheduling,” *Computers and Electronics in Agriculture*, vol. 135, 2017.

[10] Kim, Y. and Evans, R. G., “Design Criteria for PID-Controlled Drip Irrigation Systems,” *Transactions of the ASABE*, vol. 54, no. 6, 2011.

[11] Too, E. C. et al., “A Comparative Study of Fine-Tuning Deep Learning Models for Plant Disease Identification,” *Computers and Electronics in Agriculture*, vol. 161, 2019.

[12] Brahimi, M. et al., “Deep Learning for Plant Identification Using Leaf Characteristics,” *Computers and Electronics in Agriculture*, vol. 153, 2018.

[13] Picon, A. et al., “Deep Learning for Plant Disease Detection Using Leaf Images,” *Computers and Electronics in Agriculture*, vol. 153, 2018.

[14] Ferentinos, K. P., “Deep Learning Models for Plant Disease Detection and Diagnosis,” *Computers and Electronics in Agriculture*, vol. 145, 2018.

[15] Lottes, P. et al., “Real-Time Semantic Segmentation of Crop and Weed for Precision Agriculture Robots,” *Robotics and Autonomous Systems*, vol. 100, 2018.

[16] Mohanty, S. P. et al., “Using Deep Learning for Image-Based Plant Disease Detection,” *Frontiers in Plant Science*, vol. 7, 2016.